



The biogeographic history of beech trees

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ABSTRACT

Biogeographic histories cannot solely rely on cladistic reconstructions. First, a cladogram may be an insufficient representation of evolution, in particular at low taxonomic levels (here: intrageneric evolution). Diversification and speciation in plants is complex, rarely resulting in a single unambiguous tree-like genealogy. Second, the modern distribution of a monophyletic group (here: genus *Fagus*) may be highly incomplete. For instance, important areas such as Central Asia and Iceland no longer include the taxon. Third, incorporating fossil taxa, including potential ancestors of modern taxa, into a cladistic phylogeny may be intrinsically difficult. All three issues apply to the Northern Hemispheric genus *Fagus*, the beech trees. In this study, we use the exceptionally complete fossil record of *Fagus* to trace its spatiotemporal unfolding throughout the Cainozoic. Fossil taxa are placed in a phylogenetic framework following Darwinian classification by identifying common ancestries and overall patterns of similarity. These lines of evidence are then combined to put up a holistic biogeographic history of *Fagus* in the Cainozoic, which is discussed in the light of general patterns of morphological and molecular differentiation and diversification. Based on the fossil record, *Fagus* evolved in the Northern Pacific region. Earliest fossils are known both from Northeast Asia and northwestern North America (possibly extending to Axel Heiberg Island) in the Middle Eocene. The range expansion of *Fagus* into Central Asia and Europe during the Oligocene was accompanied by a first differentiation into a continental Eurasian and northern Pacific lineage. In the Miocene, *Fagus* is found throughout the Northern Hemisphere up to very high latitudes (e.g., Iceland). By the end of the Miocene several lineages had been established, including putative ancestors of modern taxa (e.g. *F. evenensis*, the ancestor of subgenus *Engleriana*). Overall, the Miocene taxa are characterized by conspicuous morphological plasticity reflecting potentially unhindered gene flow. In post-Miocene times, *Fagus* underwent range reduction and extinction in high latitudes, Central Asia and western North America. The disruption of a former homogeneous area led to the shaping of modern species. The methodological approaches used here for *Fagus* take into account the particular nature of data sets palaeobotanists are usually confronted with.

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1. Introduction

Cladistic biogeography and related divisions of historical biogeography use bifurcating trees to infer biogeographical histories for groups of organisms (Liebermann, 2000; Ebach and Humphries, 2002). While the increasing number of methods and their modifications to reconstruct the spatial diversification of taxa has caused considerable discussions in recent times (e.g., Nelson and Platnick, 1991; Ronquist, 1994; Liebermann and Eldredge, 1996; Ebach, 1999; Ebach and Humphries, 2003; Liebermann, 2003; Ebach and Morrone, 2005; Santos, 2007; Sanmartin et al., 2008) there is a general consensus that a phylogenetic framework is needed in order to make assumptions about biogeography. Over the recent years, evidence has accumulated that a single bifurcating tree may not be sufficient as phylogenetic framework for intrageneric evolution and, in general, to infer

ancestor–descendant relationships (common origin) (Denk et al., 2002; Denk et al., 2005; Grimm et al., 2007b for *Fagus*; for a more general view see Wilkinson, 1996; Doolittle, 1999; Legendre, 2000; Mayr and Bock, 2002; Felsenstein, 2004; Kunin et al., 2005; Hörandl, 2006; Huson and Bryant, 2006; Whitfield and Lockhart, 2007; Zander, 2008). This has been exemplarily shown for original texts (ancestors) and text copies made by scribes (descendants) by Spencer et al. (2004). Nevertheless, currently no method exists to take this limitation into account for biogeographic analyses.

Limitations of historical biogeographic studies due to the incompleteness of the modern distribution are generally not addressed (cf. Manos and Donoghue, 2001 and contributions in that issue; Donoghue and Moore, 2003; Lang et al., 2007) since most of these studies are exclusively based on molecular phylogenies of extant members of the various plant genera. Exceptions are the studies by Manchester and Tiffney (2001) and Xiang et al. (2006). The general notion that fossils are superfluous for the reconstruction of historical biogeography is perhaps best illustrated by Humphries and Parenti (1999, p. 48), who state that “cladistics allow us to discard the

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contention that fossils are required to reconstruct the phylogenetic or distributional history of a group". This is surprising considering the common meaning of "history" being a chronological record of events

and general problems inherent to the cladistic approach in reconstructing evolution (e.g., Mayr and Bock, 2002; Hörandl, 2006). Fossils are traces of evolutionary events through time. Probably the true

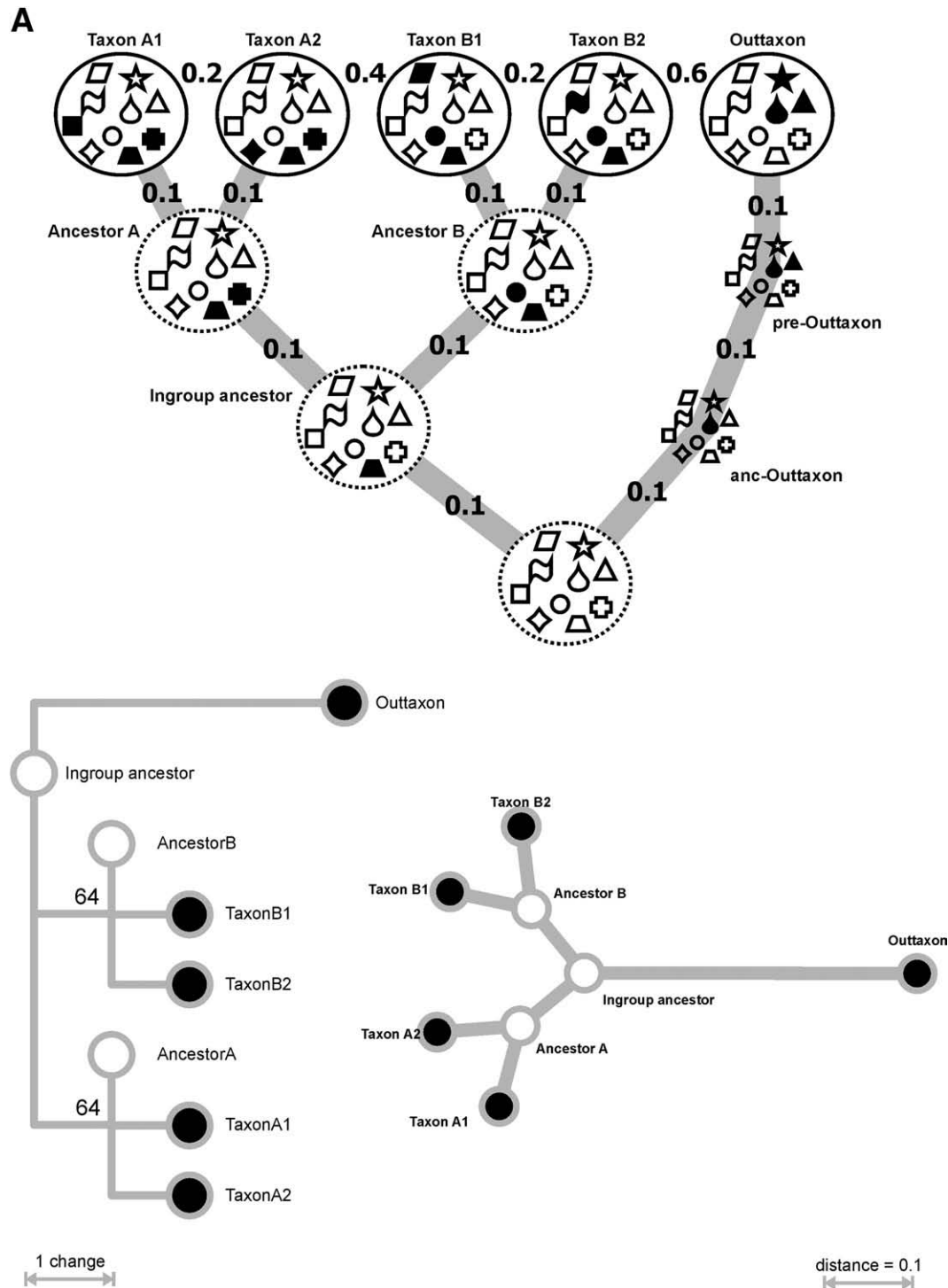


Fig. 1. 'True trees', distances and reconstructions. A, Ancestor-descendant relationships in a 'perfect' phylogeny. There are no convergent developments, and the rate of change (mutation rate) is low and constant: Each isolated lineage accumulates one derived (filled symbols) character per time interval (perfect cladogenesis). Top, the 'true tree', the actual evolution. Numbers at branches refer to the distance between two taxa (represented by circles). The distances between two extant taxa (e.g., Taxon A1 and A2), or an extant taxon and its progenitor (e.g., Taxon A1 and Ancestor A) perfectly represent their true evolutionary relationships. Bottom left, parsimony analysis results in a single most parsimonious tree (MPT), bootstrap support is indicated along the branches: Ancestors are placed as zero-length branches along a polytomy including their descendants. Bottom right, phylogenetic network (neighbour-net splits graph) based on the resulting distance matrix. The network is perfectly tree-like because all splits in the primary data are compatible. B, Ancestor-descendant relationships in an 'imperfect' phylogeny. In addition to lineage-specific characters (black), some lineages evolve independently the same derived (convergent) characters as in the outgroup or sister lineage (grey). Top, the 'true tree'; numbers at branches refer to the distance between two taxa (represented by circles). Still, the distances between two extant taxa (e.g., Taxon A1 and A2), or an extant taxon and its (direct) progenitor (e.g., Taxon A1 and Ancestor A) represent their true evolutionary relationship. Bottom left, one of six MPT, with bootstrap support indicated along branches; and the strict consensus tree of all six MPT. The reconstruction is at odds with the true phylogeny. Bottom right, phylogenetic network (neighbour-net splits graph) based on the resulting distance matrix. The network is less tree-like because some splits in the primary data are incompatible due to the convergent development of some characters. The relationship indicated between Taxon B2 and Outtaxon is a reconstruction artefact.

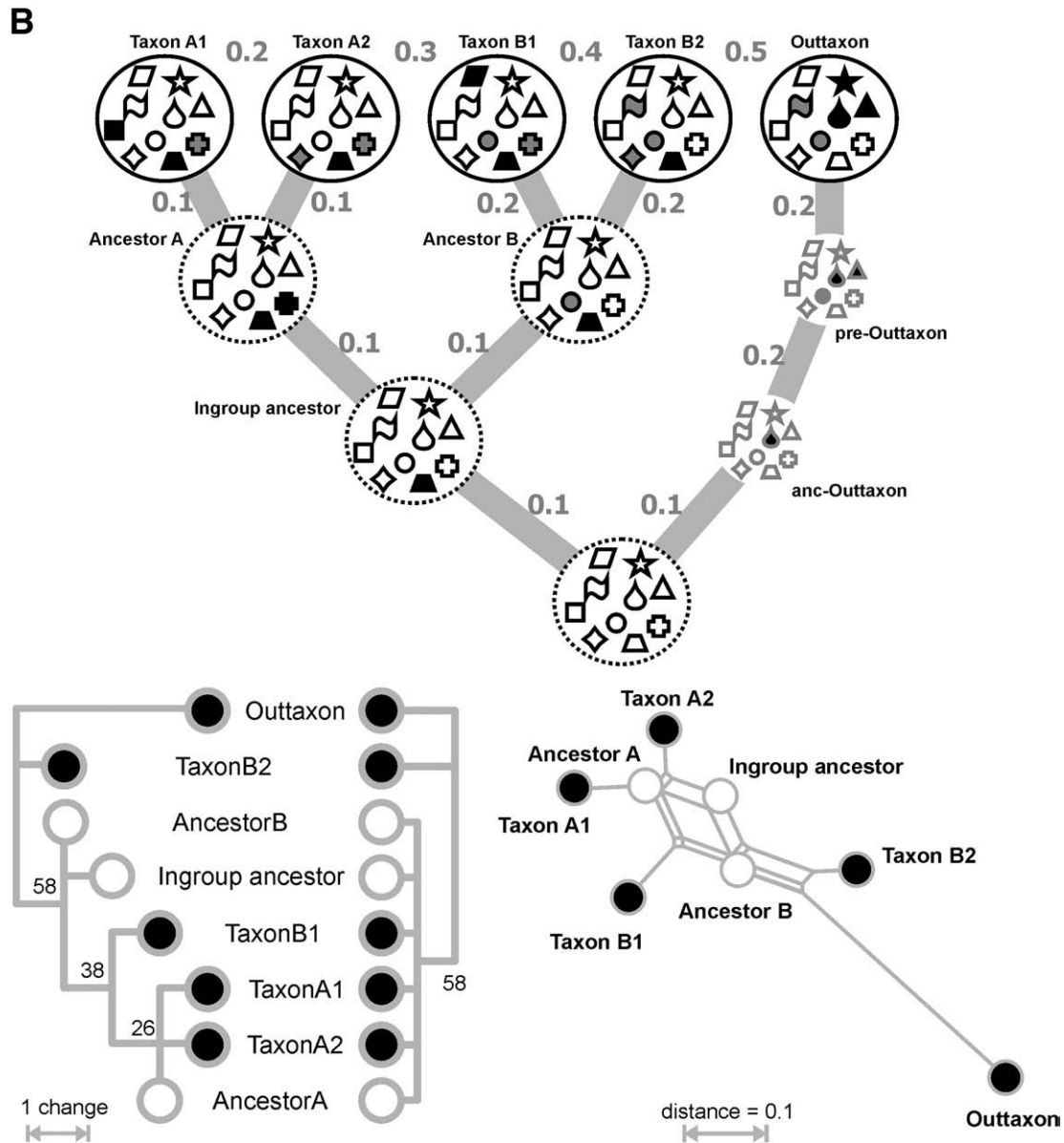


Fig. 1 (continued).

reason for statements as the one mentioned may be the various problems that arise when fossils are included in cladistic analyses (e.g. Mayr and Bock, 2002; Wiens, 2003; Felsenstein, 2004, chapter 32; Hörandl, 2007). Another unwanted feature of fossils in the context of biogeographic histories is that they consistently are found where they should not be: for instance, Lang et al. (2007), based on a molecular cladogram of extant taxa, inferred an East Asian Eocene origin for the genus *Castanea* with subsequent migration to Europe and on to eastern North America in the Palaeogene. This is highly unlikely in view of the geological history of Eurasia, with the closure of the Turgai strait only after the Eocene (Smith et al., 1994). At the same time, *Castanea* cupules are known from the Eocene of Tennessee (Crepet and Daghljan, 1980), which may indicate an initial dispersal from East Asia to North America or vice versa. Although densely studied, European sediments have not yielded *Castanea* fossils older than Early Miocene.

Felsenstein (2004) advocated that, provided a thoroughly searched fossil record, “we should not only be allowed to interpret fossils as ancestors, we should be encouraged to do so.” If fossils are interpreted as ancestors (or close relatives of ancestors), their

geographical distribution is of paramount importance to get an idea about the biogeographical history of a lineage. The only obstacle left is how to include fossils in a phylogenetic (systematic) framework. However, this is mainly a problem of cladistics (Hennigian classification, “cladification”; cladistic phylogeny; see Mayr and Bock, 2002; Hörandl, 2006, 2007). Using the principles of Darwinian classification (common descent, overall similarity; Haeckelian phylogeny; Mayr and Bock, 2002) and phylogenetic networks based on trees (consensus networks, Holland et al., 2004), phylogenetic splits (bipartition networks, e.g., Grimm et al., 2006, 2007a) and distances (e.g., neighbour-net splits graphs, Bryant and Moulton, 2002, 2004) fossils can be included in phylogenetic reconstructions (this study). For instance, Spencer et al. (2004) found that in the case of text copies, the original texts (ancestors) are generally closer to the centre of a neighbour-net splits graph (a distance-based planar phylogenetic network; Bryant and Moulton, 2002, 2004) than their copies (descendants). Spencer et al. (2004) also found that ancestor-descendant relationships are better reflected in a phylogenetic network than, for instance, in a parsimony phylogenetic tree. To illustrate this, Fig. 1 shows the placement of ancestors and their

offspring under two different evolutionary scenarios in most parsimonious trees based on the primary character matrix and in a neighbour-net splits graph based on morphological pairwise distances. Ancestors, by inducing conflicting signal, behave different than their offspring and, thus, can be recognized.

Beech trees (*Fagus* L., Fagaceae) are a small genus of flowering plants consisting of two morphologically and genetically distinct subgenera (subgenus *Fagus* and *Engleriana*; Shen, 1992; Denk, 2003; Denk et al., 2005). The genus can be considered to be monophyletic (sensu Haeckel, 1866; Mayr and Bock, 2002), a common origin of all members can be assumed (Denk et al., 2005). Members of subgenus *Engleriana* descended from a common ancestor, they form a well supported clade in phylogenetic trees (Denk, 2003; Denk et al., 2005; Fig. 2), and can be easily recognized by their overall molecular and morphological similarity (Shen, 1992; Denk, 2003; Grimm, 2003; Denk et al., 2005; Grimm et al., 2007b). Further relationships among members of subgenus *Fagus* are more complex (Fig. 2): The North American *F. grandifolia* is clearly distinct, and appears closer related to subgenus *Engleriana* than the remainder (Denk, 2003; Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008). Denk et al. (2005) concluded that the ancestor of *F. grandifolia* diverged from a Beringian-western North American population, from which also subgenus *Engleriana* evolved (see Grimm, 2003, figs 3–13, 3–15; Denk, 2004, fig. 16, Appendix A). The Japanese *F. crenata* is morphologically highly similar to the western Eurasian *F. sylvatica* (Denk, 2003), and, based on ITS data, closer related to *F. sylvatica* and/or the Chinese relict *F. lucida* (Grimm, 2003; Denk et al., 2005; Grimm et al., 2007b; represented by the red and green edges in Fig. 2) than to *F. hayatae* and *F. longipetiolata* (purple edges; Fig. 2). The latter two are morphologically distinct (Denk, 2003), but share the same ITS gene pool showing conspicuous intraindividual variability (Grimm, 2003; Denk, 2004; Denk et al., 2005; Grimm et al., 2007b). No molecular data are available for the two remaining species

F. okamotoi and *F. chienii*. Based on morphological data, *F. okamotoi* is a member of subgenus *Engleriana* and close ally of *F. japonica*; and *F. chienii* is related to other Chinese species of subgenus *Fagus*, in particular *F. lucida* (Denk, 2003; Fig. 2).

Beech trees have an exceptionally well studied fossil record extending back to the Early Cretaceous (Kolakovski, 1960; Tanai, 1974; Zetter, 1984; Kvaček and Walther, 1989, 1991; Denk, 2004; Grímsson and Denk, 2005; Manchester and Dillhoff, 2005). In earlier studies (Denk et al., 2002; Denk, 2003, 2004; Denk et al., 2005) we relied on this exceptional fossil record to discuss and to interpret patterns of ITS differentiation. In this paper we present a comprehensive biogeographic history for *Fagus* based on palaeontological data and modern species. We use the degree of morphological similarity between extant and fossil taxa, and among fossil taxa, to establish hypotheses about the timing, areas and patterns of diversification in the genus *Fagus*. This is compared to a classical dispersal-vicariance analysis (DIVA) based solely on the distribution of modern taxa.

Terminology: we use the same terminology as Mayr and Bock (2002). “Clade” and “grade” refer to topological placements in phylogenetic trees, whereas “monophyletic” refers to the common origin of a group (Haeckel, 1866). “Lineage” refers to a (monophyletic) group/succession series of fossil and/or extant taxa. “Close” and “distant” are used in an absolute manner, referring to pairwise distances based on the character matrix and the sum of branch or edge lengths, and not – as in cladistic literature – referring to the number of tree nodes (cf. Felsenstein, 2004).

2. Material and methods

Table 1 lists all extant species and the fossil taxa of *Fagus* considered for the present study, and their biogeographic and stratigraphic distribution. Analyses are based on the data matrix of Denk (2003)

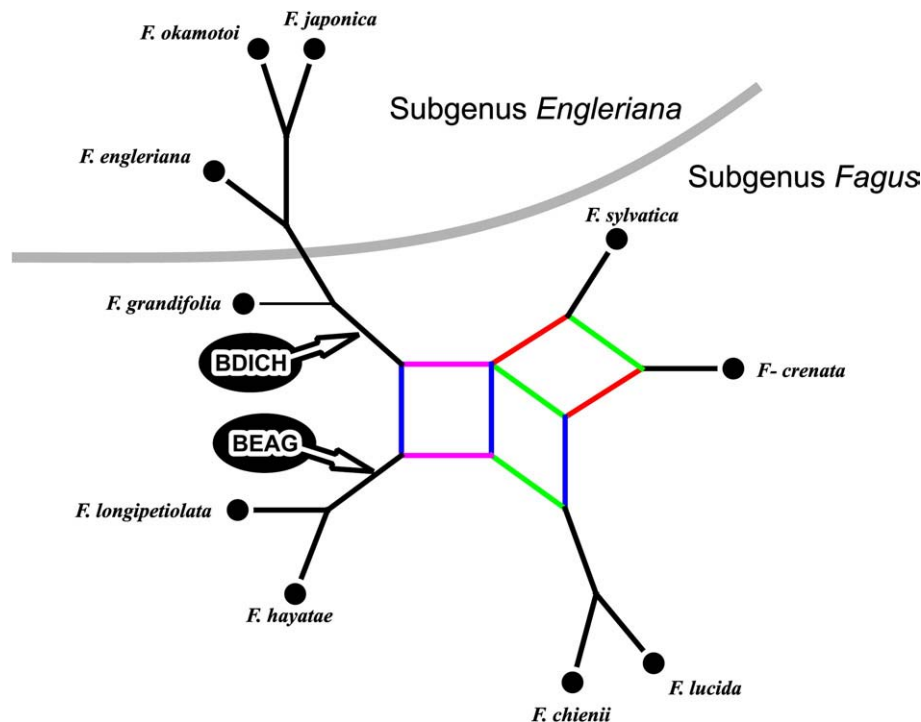


Fig. 2. Simplified representation of phylogenetic relationships in extant *Fagus*. Shown is a consensus network of three alternative (competing) cladograms, each one illustrating relationships indicated by morphological and/or molecular (ITS) data (Denk, 2003; Grimm, 2003; Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008). Two potential roots are indicated by arrows: BEAG (basal Eurasian grade) and BDICH (basal dichotomy) refer to topologies used for DIVA (see Results). The subgeneric division proposed by Shen (1992) is indicated, which has been confirmed by Denk (2003) and is in agreement with Denk et al. (2005). Coloured edges represent alternative (incongruent) relationships indicated by the underlying data: Blue edges, a phylogenetic split between the southern Chinese members of subgenus *Fagus* and the remainder is indicated by morphological data (Denk, 2003). In contrast, molecular data favour a closer relationship between *F. lucida* (+ *F. chienii*) and *F. crenata* + *F. sylvatica* (Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008). Red edges, morphological and molecular data support a common origin for *F. crenata* and *F. sylvatica* (Denk, 2003; Denk et al., 2005; Grimm et al., 2007b); in addition, green edges, patterns of intraindividual ITS variability indicate a closer relationship between *F. lucida* and *F. crenata*, but not *F. sylvatica* (Grimm et al., 2007b).

Table 1Extant and fossil species of *Fagus* used for the present study, and their spatio-temporal distribution.

	Age	Geographic range	General area	Taxon set
Fossil species of <i>Fagus</i>				
<i>Fagus langevinii</i> Manchester and Dillhoff	Eocene	British Columbia	Pacific North America	BK
<i>Fagus napanensis</i> Fotjanova	Eocene	Kamchatka, Sakhalin	Beringia	LK
<i>Fagus</i> sp. indet. ^a	Eocene	Northeastern China	Northeast Asia	Not included
<i>Fagus</i> sp. 1 Liu et al. ^b	Eocene or Oligocene	Southcentral China	South China	Not included
<i>Fagus</i> sp. 2 Liu et al. ^b	Oligocene	Northeastern China	Northeast Asia	LK
<i>Fagus pacifica</i> Chaney	Oligocene	Oregon	Pacific North America	BK
<i>Fagus uemurae</i> Tanai	Oligocene	Japan	Northeast Asia	LK
' <i>Fagus</i> ' <i>kitamiensis</i> Tanai	Oligocene	Japan	Northeast Asia	LK
<i>Fagus galbanifolia</i> Guo	Oligocene	Northeastern China	Northeast Asia	LK
<i>Fagus castaneifolia</i> Unger	Oligo-/Miocene	Central Asia to Europe	Central Asia + Europe/West Asia	WK
<i>Fagus washoensis</i> LaMotte	Miocene	Montana, Idaho	Pacific North America	BK
<i>Fagus idahoensis</i> Chaney & Axelrod	Miocene	Idaho	Pacific North America	Combined with <i>F. washoensis</i>
<i>Fagus chankaica</i> Alexeenko	Miocene	Russian Far East	Northeast Asia	LK
<i>Fagus salnikovii</i> Fotjanova	Oligo-/Miocene	Sakhalin, Kamchatka	Beringia	LK
<i>Fagus friedrichii</i> Grímsson & Denk	Middle Miocene	Iceland	Beringia + Iceland	WK
<i>Fagus evenensis</i> Chelebaeva	Miocene	Kamchatka	Beringia	BK
<i>Fagus protojaponica</i> Suzuki	Mio-/Pliocene	Japan	Northeast Asia	LK
<i>Fagus protolongipetiolata</i> Huzioka	Miocene	Korea	Northeast Asia	LK
<i>Fagus florinii</i> Huzioka & Takahashi	Mio-/Pliocene	Japan	Northeast Asia	BK
incl. <i>F. stuxbergii</i> sensu Tsukagoshi				
<i>Fagus palaeocrenata</i> Okutsu	Mio-/Pliocene	Japan	Northeast Asia	LK
<i>Fagus stuxbergii</i> (Nathorst) Tanai	Mio-/Pliocene	Japan	Northeast Asia	LK
<i>Fagus altaensis</i> Kornilova & Rajushkina	Miocene	Central Asia	Central Asia	BK
<i>Fagus haidingeri</i> Kováts	Mio-/Pliocene	Europe, Georgia (Caucasus)	Europe/West Asia	WK
<i>Fagus gussonii</i> Massalongo	Late Miocene	South Europe, Iceland	Europe/West Asia + Iceland	WK
Extant species of <i>Fagus</i>				
<i>Fagus engleriana</i> Rehd.	Subgenus <i>Engleriana</i>	South China, Ullung Do (South Korea)	South China + Northeast Asia	WK
<i>Fagus japonica</i> Maxim.		Japan	Northeast Asia	WK
<i>Fagus okamotoi</i> Shen		Japan	Northeast Asia	WK
<i>Fagus grandifolia</i> Ehrh.		Mexico, Eastern U.S. to Canada	Atlantic North America	WK
<i>Fagus hayatae</i> Pal.	Subgenus <i>Fagus</i>	South China, North Taiwan	South China	WK
<i>Fagus longipetiolata</i> Seem.		South China, North Vietnam	South China	WK
<i>Fagus lucida</i> Rehd. & Wilson		South China	South China	WK
<i>Fagus chienii</i> Cheng		North Sichuan	South China	WK
<i>Fagus crenata</i> Bl.		Japan	Northeast Asia	WK
<i>Fagus sylvatica</i> L.		Europe, Asia Minor	Europe/West Asia	WK

^a Unpublished leaves and cupules from Fushun.^b Based on two leaf fragments and a single leaf, respectively, figured as line drawings in Liu et al. (1996).

amended with fossil data (see Denk, 2004; *F. langevinii*, Manchester and Dillhoff, 2005; *F. friedrichii*, Grímsson and Denk, 2005).

We used parsimony and maximum likelihood (ML; via Bayesian inference) as optimality criterion for phylogenetic tree-building. Characters were treated as unordered in all analyses; polymorphic characters in the original matrix were recoded as additional states, e.g. "0", "0/1", and "1" were recoded as "0", "2", "1". This was done since we were mainly interested in detecting similarity; if polymorphisms are coded as e.g. "0/1", the corresponding taxon would be considered to be identical to all taxa displaying either "0" or "1". Most parsimonious trees (MPT) were inferred with a 'branch-and-bound' search with PAUP 4b10 (Swofford, 2002) using default settings (MULTREES and DEEPEST DESCENT option activated; ADDSEQ = FURTHEST). With respect to the data structure of the input matrix and the search algorithm used, no node support was generated under parsimony. Instead, the consensus network method (Holland et al., 2004) was used to visualize all competing topological alternatives provided by the MPT. The statistic relevance of the phylogenetic splits (bipartitions) found in the collection of the shortest trees was analysed using Bayesian-inferred posterior probabilities. Bayesian inference relied on the Mk + Γ model (Lewis, 2001) implemented in MrBayes 3.1 (Huelsenbeck and Ronquist, 2001; Ronquist and Huelsenbeck, 2003). MrBayes used 5,000,000 generations on 10 simultaneous (parallel) runs; each run used one cold Monte-Carlo Markov chain and no heated chain as suggested in the manual for matrices with few taxa. Bayesian-inferred posterior probabilities were based on 50,000 trees collected after all 10 runs converged.

Hamming distances (mean pairwise distances) were computed with PAUP 4b10 based on the morphological character matrix. A morphological distance (MD) of "0" indicates that two taxa are identical, whereas a MD of "1" indicates that two taxa differ in the states of all their characters. The MD matrix formed the basis for a planar phylogenetic network using the neighbour-net approach (Bryant and Moulton, 2002; Bryant and Moulton, 2004) and mirrors directly the degree of absolute similarity between two taxa (fossil and extant).

For means of discussion, we inferred a biogeographic history based on extant taxa using dispersal-vicariance analysis (DIVA, Ronquist, 1997). DIVA requires a fully resolved and rooted cladogram as input (or several alternative cladograms; Ronquist, 1996). More sophisticated (stochastic, Bayesian) methods than parsimony could not be used to reconstruct biogeographic scenarios due to the lack of well defined genetic branch lengths.

We used two topologies for DIVA based on previous studies by Denk (2003), Denk et al. (2005) and Grimm et al. (2007b; bold edges in Fig. 2) that only differ in the placement of the root. In the 'basal Eurasian grade' (BEAG) topology, *F. hayatae* and *F. longipetiolata* form a sister clade to the remaining species (Denk, 2003; Denk et al., 2005); subgenus *Fagus* is then paraphyletic. This topology represents the best-possible, however oversimplified, tree-like representation of phylogenetic relationships in *Fagus* based on morphology, ITS sequence data and patterns of intraindividual ITS variability. In the 'basal dichotomy' (BDICH) topology, the root is placed according to evidence from the fossil record, indicating an initial divergence between the Eurasian (leading to the Eurasian members of subgenus

Fagus) and Pacific lineages (*F. grandifolia*, subgenus *Engleriana*; Grimm, 2003, Figs. 3–13,3–15; Denk et al., 2005).

The following areas are relevant to the study of biogeographic patterns in *Fagus*: Europe and Western Asia (EWA); Central Asia (CAS); Southwest China including Taiwan and northern Vietnam (SCN); north-eastern Asia including north-eastern China and adjacent regions of Korea, Russia, and Japan (NEA); the northern Pacific region or “Beringia” including Sakhalin, Kamchatka, and Alaska (BER); Pacific North America (PNA), Mexico (MEX), and eastern (Atlantic) North America (ANA); Iceland (ISL). Of these areas CAS, BER, PNA and ISL no longer include the genus *Fagus*; and modern *Fagus* in NEA is restricted to Japan and Ullung Do island (off Korea; Table 1). BER (Beringia; Russian Far East to Alaska) represents a homogeneous Early Cainozoic biota characterized by numerous endemic taxa not found in other parts of East Asia (NEA, SCN) but in western North America (PNA; for example, *Davidia*, *Camptotheca*, Manchester et al., 1999; *Dipteronia*, McClain and Manchester, 2001; *Celtis*, Manchester et al., 2002; *Ulmus*, Feng et al., 2003; *Carpinus*, *Corylus*, Pigg et al., 2003; *Koelreuteria*, Wang, in press; *Fagus*, Denk, 2004; Denk et al., 2005; this study).

Modern beeches are, with the exception of *F. grandifolia* (ANA, MEX), restricted to geographic regions characterized by Cfb and Cfa climates according to Köppen's system (warm temperate, fully humid climates; Köppen and Geiger, 1928; Köppen, 1936; Kottke et al., 2006; Peel et al., 2007). The extant centre of diversity for *Fagus* is southern China including Taiwan and northern Vietnam (SCN; Shen, 1992).

3. Results

3.1. Fossil record of *Fagus*

Earliest unequivocal fossils of the genus *Fagus* (leaves, cupules, nuts, and pollen) are from the late Early Eocene (± 50 Ma; *Fagus langevinii* [LAN] Manchester and Dillhoff, 2005; see Denk and Dillhoff,

2005 for age determination; Table 1, Fig. 3) and Middle Eocene (nuts; *F. schofieldii* Mindell et al., 2009) of western North America. Dispersed pollen has been reported and figured from the Middle Eocene of Axel Heiberg Island (McIntyre, 1991; Richter and LePage, 2005). Soon after, the genus made its appearance in the Late Eocene of the Russian Far East (*F. napanensis* [NAP] Fotjanova, 1982) and in the Middle to Late Eocene of north-eastern China (*Fagus* sp. [?SP], leaves and cupules from Fushun, unpublished data; Table 1, Fig. 3). The late Eocene to Early Oligocene *Fagus* sp. 1 (not shown) from south-central China (Liu et al., 1996) is based on two leaf fragments that cannot unambiguously be placed within *Fagus*. The Oligocene was characterized by radiations in north-eastern Asia, Beringia, and Pacific North America, and by range expansions accompanied by differentiation in Eurasia (Fig. 3). In western North America, the Early Oligocene *F. pacifica* [PAC] Chaney (1927) occurred at about the same time as two species from Japan (*F. uemurae* [UEM], *F. kitamiensis* [KIT] Tanai, 1995; Table 1). From the Oligocene of north-eastern China, *F. galbanifolia* [GAL] Guo (Liu et al., 1996) including *F. koraica* sensu Guo and Zhang (2002; Table 1, Fig. 3) resemble *F. pacifica*. *Fagus* sp. 2 [SP2] (Liu et al., 1996) from the same area and time is based on a single small entire-margined leaf specimen. In Eurasia, the genus extended westwards in the Early Oligocene following the northern shores of the Paratethyan Sea and migrating into Europe after the closure of the Turgai Sea ca. 37 Ma (Smith et al., 1994; Akhmetiev and Reshetov, 1996; Schmid, 2000; Table 1, Fig. 3). Hence, in Central and West Asia and Europe a single species, *F. castaneifolia* [CAS] Unger was widespread from the late Early Oligocene to the Early/Middle Miocene (Denk, 2002, 2004; Table 1, Fig. 3).

In terms of the number of named species, *Fagus* reached its peak diversity in the Miocene (Table 1, Fig. 3). In western North America, *F. washoensis* La Motte (1936) and the closely related *F. idahoensis* Chaney and Axelrod (1959) [I/W] occurred from the Early to the Late Miocene and may represent a lineage leading to the modern *F.*

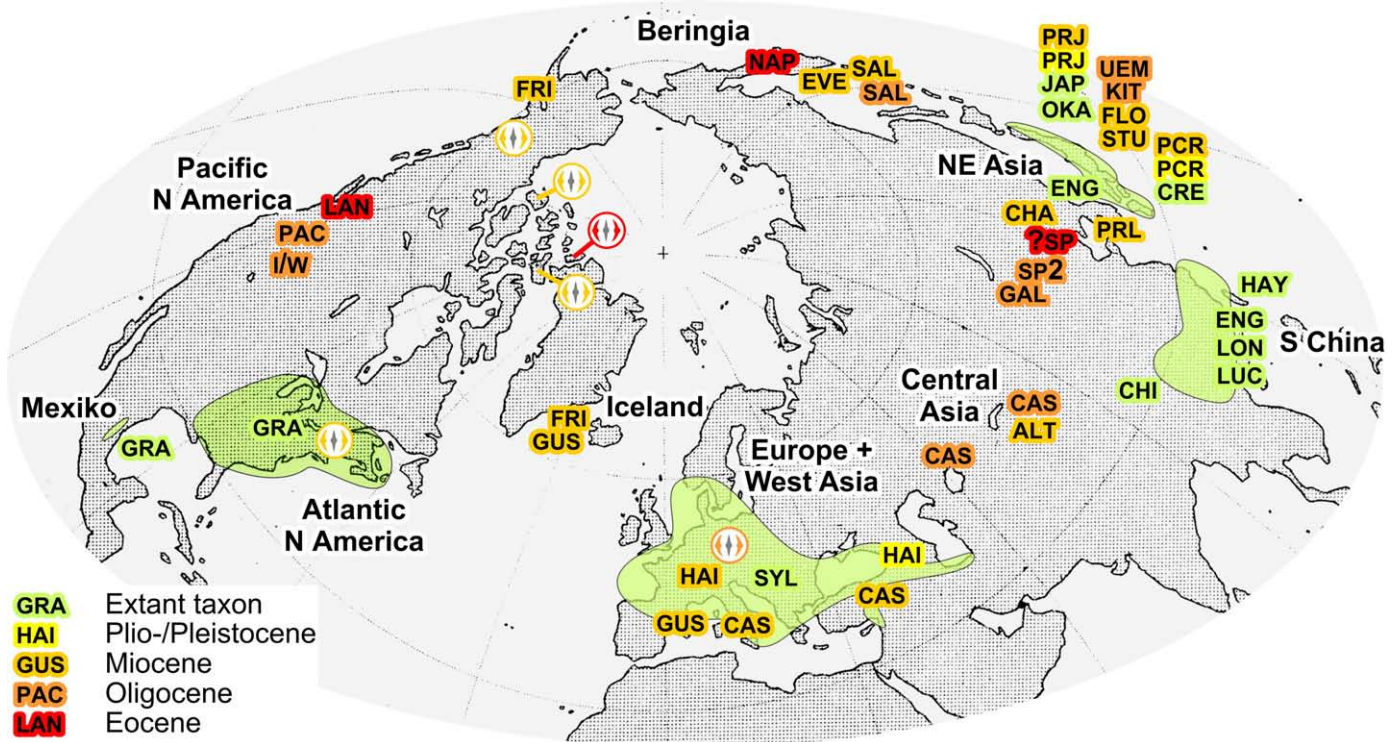


Fig. 3. Historical biogeography of *Fagus*. Modern distribution indicated by green shading. Taxon names are abbreviated: ?SP, *Fagus* sp. from Fushun, northeastern China; ALT, *F. altaensis*; CAS, *F. castaneifolia*; CHA, *F. chankaica*; CHI, *F. chienii*; CRE, *F. crenata*; ENG, *F. engleriana*; EVE, *F. evenensis*; FLO, *F. florinii*; FRI, *F. friedrichii*; GAL, *F. galbanifolia*; GRA, *F. grandifolia*; GUS, *F. gussonei*; HAI, *F. haidingeri*; HAY, *F. hayatae*; I/W, *F. idahoensis-washoensis*; JAP, *F. japonica*; KIT, *F. kitamiensis*; LAN, *F. langevinii*; LON, *F. longipetiolata*; LUC, *F. lucida*; NAP, *F. napanensis*; OKA, *F. okamotoi*; PAC, *F. pacifica*; PCR, *F. palaeocrenata*; PRJ, *F. protojaponica*; PRL, *F. protolongipetiolata*; SAL, *F. salnikovii*; SP2, *Fagus* sp. 2; STU, *F. stuxbergii*; UEM, *F. uemurae*. Tricolporate pollen symbols indicate dispersed *Fagus* pollen.

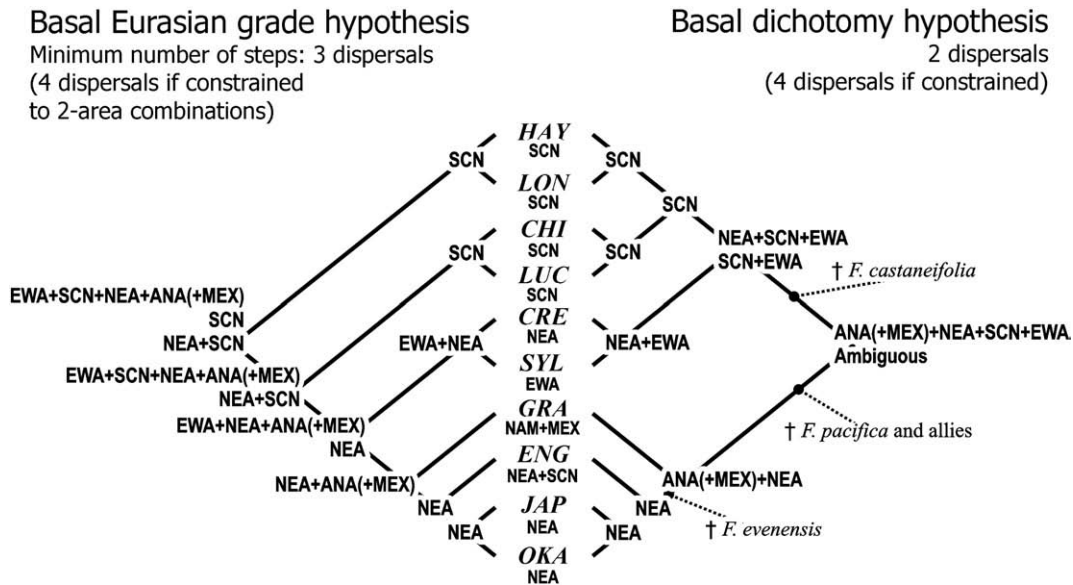


Fig. 4. DIVA reconstructions using the modern distribution of *Fagus*. For both alternative placements of the *Fagus* root ('basal Eurasian grade', BEAG, left, and 'basal dichotomy', BDICH, right, topology), the DIVA reconstruction is inconclusive considering the biogeographic history of *Fagus*. Abbreviations of taxa: CHI, *F. chienii*; CRE, *F. crenata*; ENG, *F. engleriana*; GRA, *F. grandifolia*; HAY, *F. hayatae*; LON, *F. longipetiolata*; LUC, *F. lucida*; OKA, *F. okamotoi*. Abbreviation of areas: ANA, Atlantic North America; BER, Beringia; CAS, Central Asia; EWA, Europe and Western Asia; ISL, Iceland; MEX, Mexico; NEA, Northeast Asia; PNA, Pacific North America; SCN, Southern China.

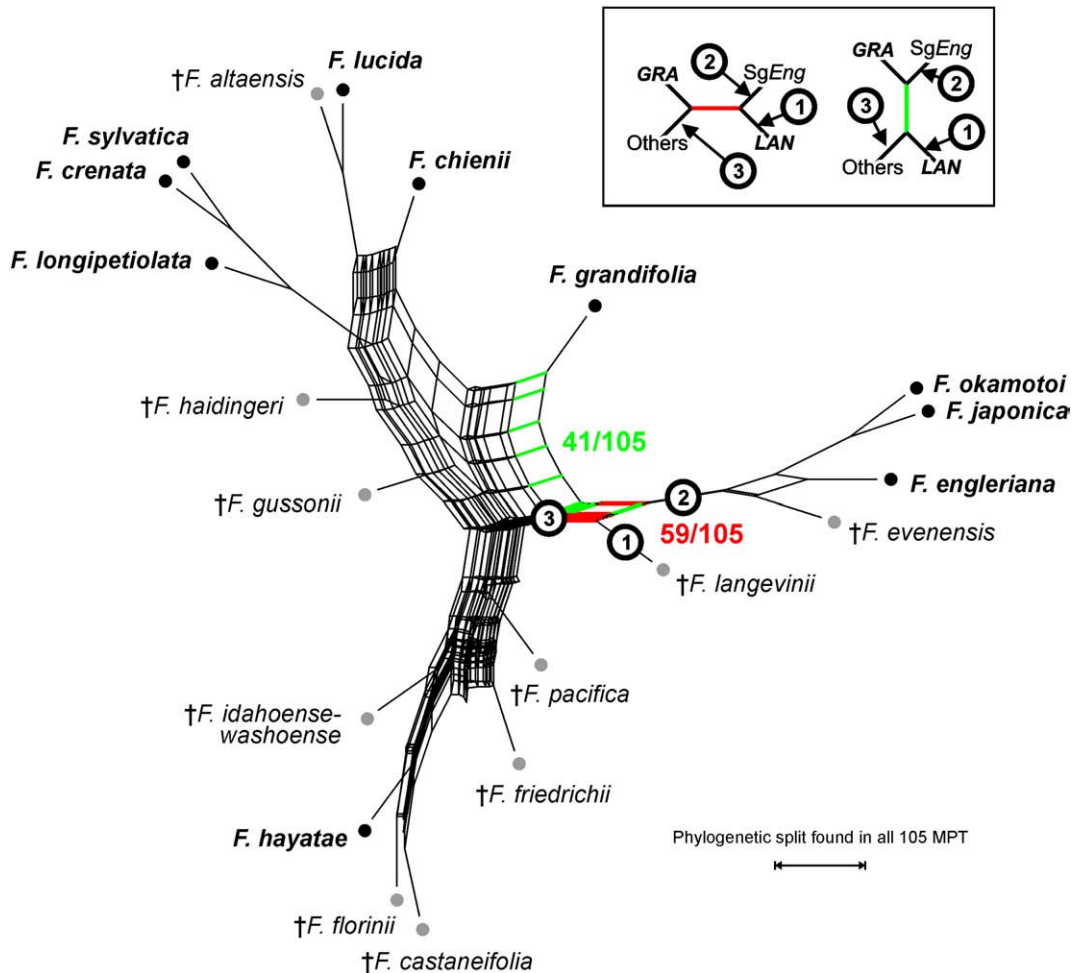


Fig. 5. Consensus network of all most parsimonious trees. Extant taxa in bold font, extinct taxa in normal font. The edge length is proportional to the number of MPT showing the respective phylogenetic splits. Numbers indicate three potential positions of the root, the common ancestor of all *Fagus*: ① *F. langevinii* as "common ancestor"; ② Shen's (1992) root between subgenus *Engleriana* and *Fagus*; ③ BDICH-root. Red, green: Alternative topologies considering the position of *F. grandifolia*, *F. langevinii* and subgenus *Engleriana* (including *F. evenensis*). Inset: Unrooted trees corresponding to the two (red, green) alternatives.

grandifolia [GRA] (Denk, 2004). The fossil record of *Fagus* in eastern North America is sparse. However, pollen from the Early Miocene Brandon Lignite can be assigned to *Fagus* with certainty (Traverse, 1994; Fig. 3). Also the Early to Late Miocene *F. chankaica* [CHA] Alexeenko (in Krassilov and Alexeenko, 1977) from the Russian Far East (Primorje) resembled *F. grandifolia* (Iljinskaya, 1982).

By contrast, *F. evenensis* [EVE] Chelebaeva (1980) from the Early to Middle Miocene of Kamchatka and Sakhalin strongly matched members of the modern subgenus *Engleriana* Shen (1992) and clearly differed from other coeval *Fagus* species. *Fagus evenensis* differs from coeval species by its densely spaced tertiary and secondary veins (Chelebaeva, 1980, 1982) and the constantly entire margined leaves and brochidodromous to pseudocraspedodromous type of secondary venation along with a distinct intrasecondary margin region. Such a character combination is exclusively found in the extant members of the subgenus *Engleriana* (Denk, 2004). Later, the lineage probably leading to the modern subgenus *Engleriana* was represented by the Late Miocene to Pleistocene *F. protojaponica* [PRJ] K. Suzuki (1959) from Japan (Table 1, Fig. 3).

A slightly older species from the latest Oligocene to the Early Miocene of Kamchatka and Sakhalin, *F. salnikovii* [SAL] Fotjanova (1988), was morphologically distinct from *F. evenensis* (Grímsson and Denk, 2005). This species may have belonged to a lineage, comprising high latitude species from the Russian Far East (*F. salnikovii*), Alaska, and Iceland (*F. friedrichii* [FRI] Grímsson and Denk, 2005; Table 1, Fig. 3). Dispersed *Fagus* pollen is also known from the Middle Miocene of Alaskan–Yukon border and Banks Island (Northwest Territories; Fyles et al., 1994; White and Ager, 1994), and possibly from the Early Miocene of Devon Island (Nunavut; Whitlock and Dawson, 1990; Fig. 3).

During the later Neogene, *Fagus* was rare in north-eastern China and most of the diversity occurred in Japan and Korea facing the Pacific Ocean (Table 1, Fig. 3). From Korea, *F. protolongipetiolata* [PRL] (incl. *F. koraica* auct. and *F. uotanii*) was described (Huzioka, 1951, 1972). From Japan, *F. florinii* [FLO] Huzioka and Takahashi (1973; including *F. stuxbergii* sensu Tsukagoshi, 1998; Denk, 2004), *F. palaeocrenata* [PCR]

Okutsu (1955), and *F. stuxbergii* [STU] (Nathorst) Tanai (Nathorst, 1883; Tanai, 1976) were described. *Fagus altaensis* [ALT] Kornilova and Rayushkina (1968; Rayushkina, 1982) was restricted to the Miocene of Central Asia.

In Central Asia and Europe *F. castaneifolia* persisted into the Early Miocene (Denk, 2002, 2004), while other taxa occurred for the first time (*F. haidingeri* [HAI] Kováts, 1856 sensu Knobloch, 1969, *F. gussonii* [GUS] Massalongo, 1858 sensu Knobloch and Velitzelos, 1986; Table 1). It is noteworthy that the predominantly South and south-western European *F. gussonii* has been documented from Late Miocene sediments in Iceland (Grímsson and Denk, 2005; Table 1, Fig. 3). During the Pliocene, Pleistocene, and Holocene the modern ranges of *Fagus* species were shaped. *Fagus* disappeared from western North America (Smiley, 1963), whereas *F. grandifolia* persisted in the South and East. In north-eastern Asia subgenus *Engleriana* [ENG,JAP,OKA] extended its range to Central and southern China. The timing of the emergence of *F. lucida* [LUC], *F. chienii* [CHI], *F. hayatae* [HAY], and *F. longipetiolata* [LON] and their migration to Southern China and adjacent Vietnam and Taiwan is obscure. *Fagus sylvatica* [SYL] and *F. crenata* [CRE] represent survivors of their western Eurasian and north-eastern Asian ancestors.

3.2. Biogeographic history based on present-day distribution

Fig. 4 shows DIVA-reconstructed ancestral areas for *Fagus* based on two topologies that differ only in the position of the *Fagus* root. The basal Eurasian grade (BEAG) topology assumes that subgenus *Fagus* is 'paraphyletic', whereas the basal dichotomy (BDICH) topology assumes that the first divergence was between the (continental) Eurasian taxa and the North American *F. grandifolia* plus Northeast Asian taxa of subgenus *Engleriana*. The BEAG topology reflects the (cladistic) synopsis of morphological and molecular phylograms (Denk, 2003; Denk et al., 2005). It represents the best-possible tree-like image of the evolutionary unfolding of *Fagus* (see Fig. 2 for a comprehensive representation). However, the outgroup-suggested

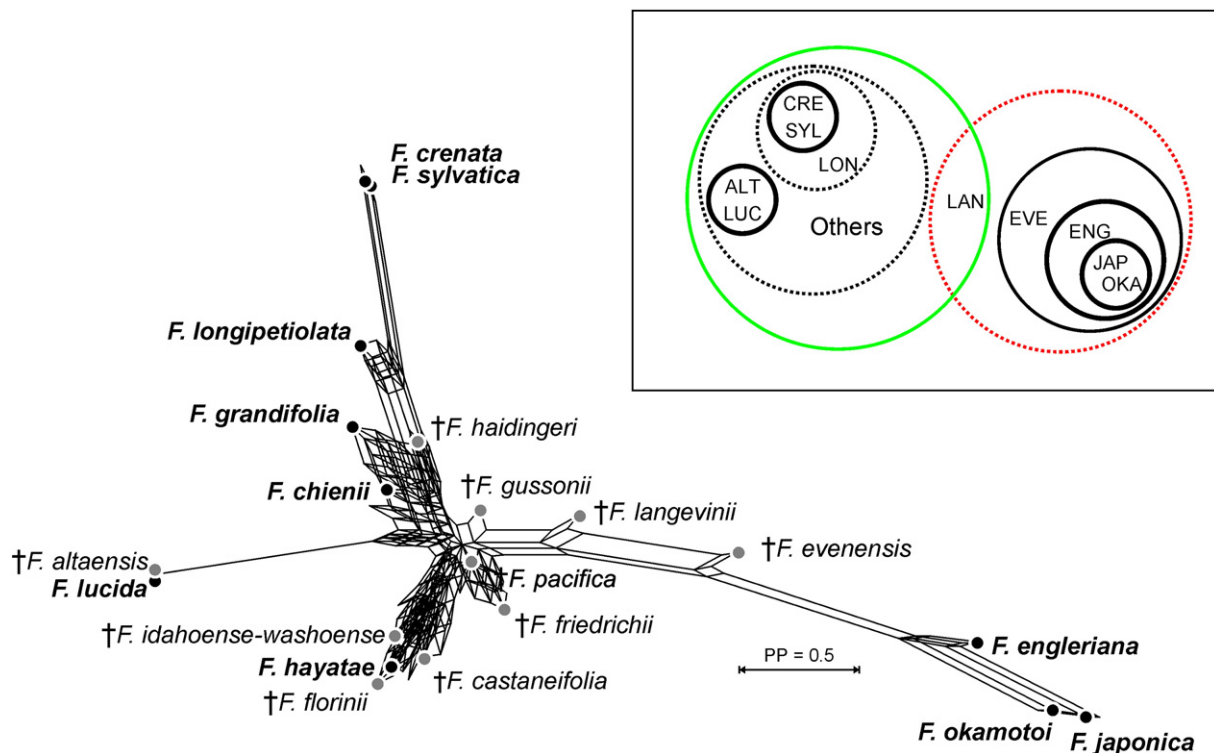
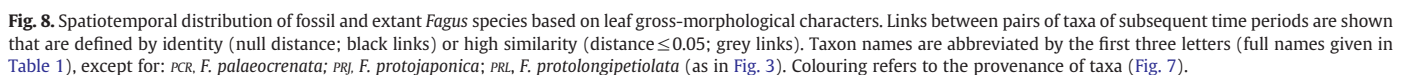


Fig. 6. Bipartition network based on Bayesian posterior probabilities (PP) of alternative phylogenetic splits. Edge lengths are proportional to the PP of the respective bipartition. Inset: Diagram summarizing the support of potential monophyletic groups (solid: PP ≥ 0.8; dotted: PP = 0.5–0.8). Colours refer to the two alternatives shown in Fig. 5.



Asia to Europe, the other by *F. pacifica* and allies (see below) and the slightly younger *F. evenensis* in western North America and the Russian Far East (Figs. 3 and 4). Hence, the BDICH topology has been



chosen to illustrate the effect of the corresponding alternative root on the reconstruction.

Independent of the rooting, DIVA is inconclusive considering the range of the common ancestor of *Fagus*. Moreover, the reconstructed ancestral areas are often difficult to reconcile with the actual fossil record and the tectonic history (see Discussion). As shown (Fig. 4) two markedly different scenarios for the origin of *Fagus*, an ubiquitous Northern Hemispheric ancestor vs. an Eastern Asian or Southern Chinese ancestor, require the same number of steps (three dispersals) based on the BEAG topology; hence, the details are not shown.

3.3. Placing fossils in a phylogenetic framework

Of the 55 characters in the morphological matrix, 26 are constant for the ingroup of extant and fossil *Fagus*. In total we included 22 fossil taxa; *F. washoensis* and *F. idahoensis* were combined for the analyses. Of these, 12 represent leaf fossils with very limited information (taxon set LK ["little known"]; Table 1; see Electronic Supplement for character matrix, character states and character coding, tabulated and in NEXUS format). For ten taxa more than half of the characters (both reproductive and foliar characters) are available and in the case of the four best known fossils more than two thirds of the characters are known (taxon sets BK, "better known", and WK, "well known"; Table 1; Electronic Supplement). Phylogenetic inferences (trees and networks) were performed excluding taxon set LK. These taxa were included in the investigation of general patterns of similarity between fossil and extant *Fagus* spp.

Parsimony analysis (branch-and-bound search) of the matrix excluding LK- and outgroup taxa resulted in 105 most parsimonious trees (MPT; 83 steps; CI = 0.51; RI = 0.62). A consensus network of all MPT is shown in Fig. 5. Fig. 6 shows a bipartition network illustrating the statistical support, the Bayesian posterior probability, of alternative phylogenetic splits (bipartitions) under maximum likelihood. Some relationships (bipartitions) are unambiguously indicated among the MPT (putative roots given in Fig. 2): (i) *F. altaensis* (Miocene; Central Asia) is placed as sister to *F. lucida* (Extant; Southern China); this placement received high support (Fig. 6). (ii) The extant *F. crenata* and *F. sylvatica* represented a well supported sister pair. (iii) In 64 of the 105 MPT subgenus *Engleriana* (East Asia; NEA + SCN) formed a clade; in the 41 remaining MPT, the fossil taxon *F. evenensis* (Beringia) is nested among the extant members of subgenus *Engleriana* (Fig. 5). The 64 MPT differ from each other by either placing *F. grandifolia* (extant; Atlantic North America) or *F. evenensis* as sister taxon to the East Asian subgenus *Engleriana* (Fig. 5). Notably, a *F. evenensis*-subgenus *Engleriana* clade received the highest support of all topological alternatives (Fig. 6).

Aside from *F. altaensis* and *F. evenensis*, the position of other fossil taxa remained ambiguous using phylogenetic tree-building under parsimony and ML. The MPT comprised numerous incongruent topologies (Fig. 5), and corresponding bipartitions of the taxon set received generally low support (distorted 'Eiffel Tower' pattern in central part of Fig. 6).

Considering the potentially earliest phylogenetic split(s), possibly marked by the position of *F. langevinii* (Eocene, Pacific North America), the controversial signal from the phylogenetic tree-building inferences can be further understood using a phylogenetic network based on the pairwise (Hamming) morphological distances (Fig. 7). *Fagus langevinii*, along with *F. evenensis*, is the fossil taxon that is most similar to the members of subgenus *Engleriana*. It is also more similar to *F. evenensis* in its characters than most other fossils of *Fagus*. This

similarity is based on leaf index and the exclusive presence of antler-like appendages on cupules in *F. langevinii*, *F. evenensis*, and *F. engleriana*. In addition, pollen of *F. langevinii* is similar to that of subgenus *Engleriana* in size and colpus length. However, *F. langevinii* is closer to the extant members of subgenus *Fagus* than *F. evenensis*. The latter differs from *F. langevinii* and subgenus *Fagus* by the general absence of trullate leaves, areoles without veinlets, and the strictly entire leaf margin. This is well illustrated by the placement of both taxa in Fig. 7. *Fagus grandifolia* is nested within subgenus *Fagus*, and clearly distant to subgenus *Engleriana*. Moreover, the network (Fig. 7) shows that the fossil taxa of *Fagus* with exception of *F. altaensis* are generally less distinct to each other than the extant taxa (cf. Fig. 1).

The shown phylogenetic reconstructions (Figs. 5–7) allow general statements on reconstructions of biogeographic history: The actual distribution of morphologically similar and potentially related taxa (extant and fossil) is often at odds with "ancestral areas" reconstructed by the DIVA analysis of extant taxa (Figs. 4–7). Even a somewhat ambiguous fossil record (Figs. 5–7) adds substantial information on the biogeographic history of a lineage.

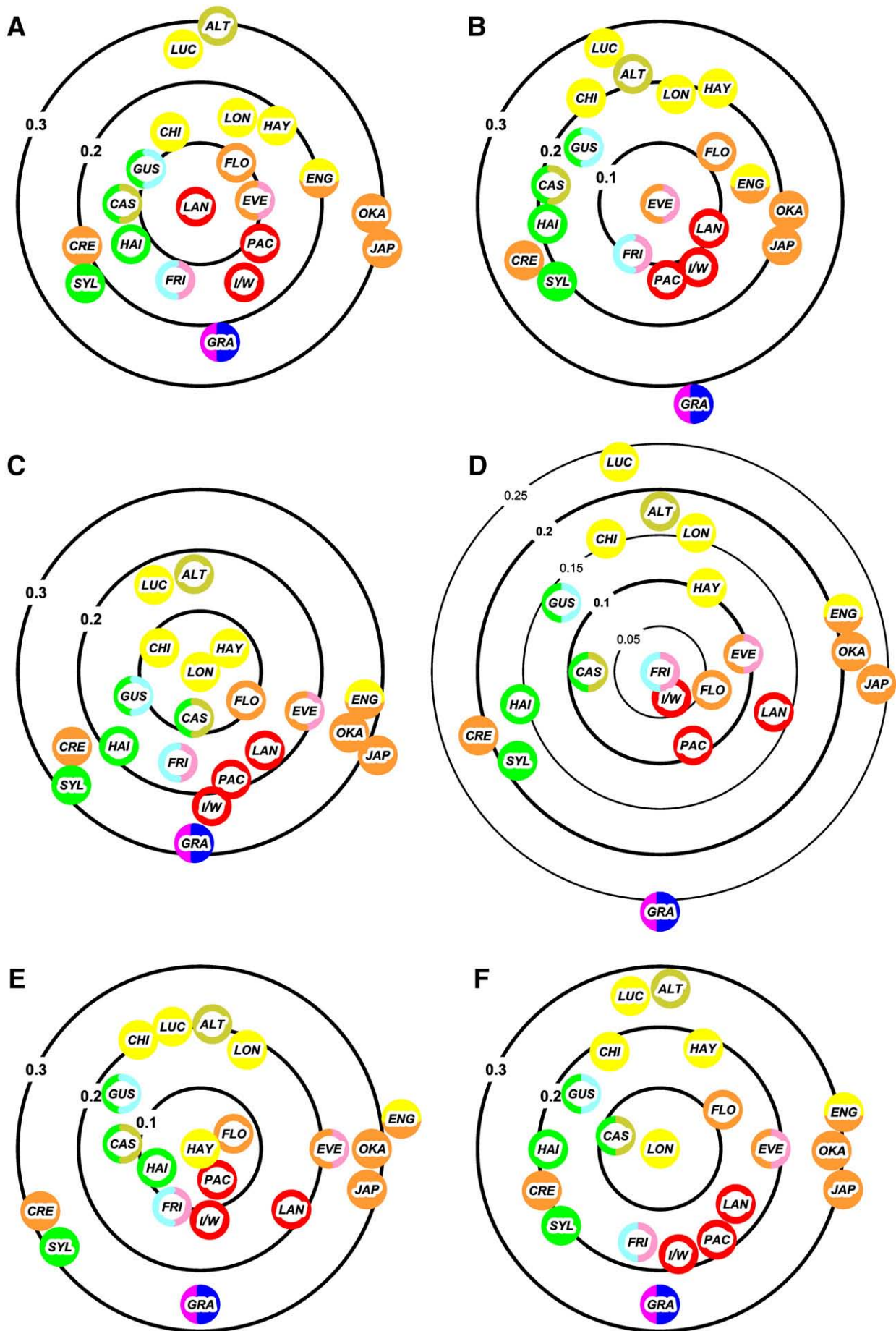
3.4. Patterns of similarity in a spatiotemporal framework

Phylogenetic trees and networks (to a lesser degree; Fig. 1) encounter limitations when placing fossils in an evolutionary and spatiotemporal context (Figs. 5–7). Therefore, we investigated patterns of (overall) similarity among fossil taxa and between fossil taxa and their modern relatives by establishing and analysing matrices of pairwise distances. The distance given in each cell of such a matrix mirrors the absolute degree of similarity between two taxa independent of any reconstruction. A distance of "0" indicates that two taxa are indistinguishable in the recorded characters, whereas a distance of "1" indicates that two taxa differ in all characters recorded. Pairwise distances were based on reduced matrices, which considered only characters available for all (most) of the included taxa.

3.5. Distribution patterns based on gross-morphological characters

Fig. 8 shows links defined by 100% and 95% identity between fossils of different geological epochs and modern taxa. Only leaf gross-morphological characters that can be scored for all fossil taxa were considered (character set MP, "maximum preservation", defined in NEXUS file; Electronic Supplement includes the underlying distance matrix). As such it reflects character evolution rather than relationships among the taxa (cf. Discussion). Several observations (Roman numbers in Fig. 8) are important for the discussion of fossils in a biogeographic context. The character combinations of most modern taxa are different from those found in fossil taxa. The only exception is *F. longipetiolata*, which is indistinguishable from the fossil *F. castaneifolia* (Fig. 8 I). *Fagus chienii*, *F. hayatae*, and *F. grandifolia* differ only slightly from Neogene fossils (II). Among Miocene taxa, *F. altaensis*, *F. palaeocrenata*, and *F. haidingeri* are distinct in their character combinations (III); both to modern (potential descendants) and to Palaeogene taxa (potential ancestors). *Fagus castaneifolia*, representing the oldest record of *Fagus* in Central Asia to Europe, is the only taxon of these two regions that was relatively similar in gross leaf morphology to earlier (*F. langevinii*, Eocene, Pacific North America) and later (Miocene to Recent) *Fagus* species (IV). Morphotypes similar in their gross-morphology to the leaf of *F. langevinii* from the Eocene of western North America persist until the Miocene of Northeast Asia (V). Most Oligocene and Miocene taxa of Northeast

Fig. 9. Degree of similarity of selected fossil taxa to extinct and extant *Fagus* species. Taxa in focus are placed as a 'sun' in the centre. 'Orbits' of other taxa reflect their pairwise distance to the focus taxon (taxon sets BK, WK; character sets OP and MP); the full table can be found in the Electronic Supplement. A, *Fagus langevinii* (Eocene, Pacific North America), the oldest macrofossil record of the genus; B, *Fagus evenensis* (Miocene, Beringia), the possible progenitor of the modern subgenus *Engleriana* (see Figs. 4–6); C, *Fagus castaneifolia* (Oligocene to Miocene, Europe to Central Asia), the prototype of a continental Eurasian *Fagus* lineage; D, *Fagus friedrichii* (Miocene of Beringia and Iceland), an extinct high-latitude member of *Fagus*. Colouring refers to the provenance of the taxa (see Fig. 7). E, *Fagus hayatae*, the least derived modern taxon (South-Central China to Taiwan). F, *Fagus longipetiolata*, the closest extant relative of *F. hayatae* (Central China to North Vietnam).



Asia, Beringia, and Pacific North America show a highly similar combination of characters (VI), and this general type extends to Iceland (*F. friedrichii*; see also Fig. 9D). A third group (VII) including the potential ancestor of subgenus *Engleriana*, *F. evenensis*, is distinct from coeval continental Eurasian types (Europe to Northeast Asia; III, IV) and the major, ubiquitous type (Northeast Asia to Iceland; V, VI). This group is restricted to the western part of the Northern Pacific region (Northeast Asia to Beringia), and stratigraphically spans from the dawn of *Fagus* until subrecent times (*F. protojaponica* extends to the Pleistocene of Japan).

In conclusion, despite their obvious variability and probably high susceptibility to parallel evolution, gross-morphological characters of fossil and extant leaves do not appear to be arbitrarily distributed through space and time, but mirror aspects of biogeographic history (Fig. 3) and phylogenetic context (Figs. 5–7) of *Fagus*.

3.6. Distance patterns among fossil and modern taxa of *Fagus*

Pairwise distances range between 0.03 (*F. friedrichii*, Miocene of Beringia and Iceland, and coeval *F. idahoensis-washoensis*, Pacific North America) and 0.39 (*F. engleriana*, Southern China and Northeast Asia, and *F. lucida*, Southern China; taxon sets WK + BK; character sets MP, mostly preserved, and OP, often preserved; data and distance table provided in the Electronic Supplement). In general, the fossil taxa are more similar to each other than the modern taxa (Figs. 7, 9). The taxon most similar to a modern species is often a fossil taxon, with the exception of the three taxa of subgenus *Engleriana* (*F. engleriana*, *F. japonica*, and *F. okamotoi*) and the disjunct siblings *F. crenata* and *F. sylvatica*.

It is not surprising that the ancestor of a modern lineage, such as subgenus *Engleriana*, should be more similar to the primary ancestor of the entire genus than its offspring (cf. Fig. 1), the extant members of subgenus *Engleriana*. Moreover, under the assumption that at least for some characters the rate of change is low enough to prevent extreme homoplasy and parallelism, the ancestors of modern lineages should be more similar to each other than are their offspring (cf. Fig. 1). Some examples are given in Fig. 9, all of which are important for the discussion (see Electronic Supplement for remaining taxa).

The oldest fossil assigned to the genus *Fagus*, *F. langevinii* (Eocene, Pacific North America; Fig. 9A) is more distant to all extant taxa than to any fossil taxon with the exception of *F. altaensis* from the Miocene of Central Asia. The Oligocene range extension of *Fagus* was accompanied by substantial radiation expressed in the high number of fossil taxa and their morphological distinctness. In addition, Oligocene and Miocene fossil taxa from western North America are less similar to *F. langevinii* than Miocene taxa from Northeast Asia (*F. evenensis*, *F. florinii*) or Europe and Iceland (*F. gussonii*). This is in analogy to gross morphological differentiation patterns (Fig. 8).

Also *Fagus evenensis* (Miocene, Beringia; Fig. 9B) is different from all extant taxa and more similar to fossil taxa. Among the most similar taxa are the Palaeogene *F. pacifica* and *F. langevinii* from the adjacent Pacific North America. The remarkable exception is the modern *F. engleriana* (Southern China and Ullung Do, Korea), which is more similar to *F. evenensis* than the fossil taxa *F. castaneifolia*, *F. gussonii*, and *F. haidingeri* from Europe, Central Asia, and Iceland.

Fagus castaneifolia (Oligocene to Miocene, Europe to Central Asia), on the other hand, is already distant to the western North American fossils and more similar to Miocene fossils from other regions (Iceland, Europe and West Asia, Northeast Asia; Fig. 9C). The extant

F. chienii, *F. hayatae*, and *F. longipetiolata* from Southern China are as similar to *F. castaneifolia* as the latter is to the slightly older *F. pacifica*.

A number of Miocene taxa of *Fagus* are morphologically indifferent. One example is *F. friedrichii* from Beringia and Iceland (Fig. 9D). This high-latitude species is highly similar to coeval but also to older fossils, while it is increasingly distinct from modern members of the genus with the exception of *F. hayatae*.

Among the extant taxa, *F. hayatae* (South-Central China to Taiwan) is most similar to a number of fossil species including the Oligocene *F. pacifica* and the Miocene to latest Pliocene *F. florinii* (Japan) and *F. haidingeri* (western Eurasia; Fig. 9E). Its closest extant relative, *F. longipetiolata* is generally less similar to the fossil taxa except for the widespread Eurasian Oligo-/Miocene *F. castaneifolia* (Fig. 9F).

4. Discussion

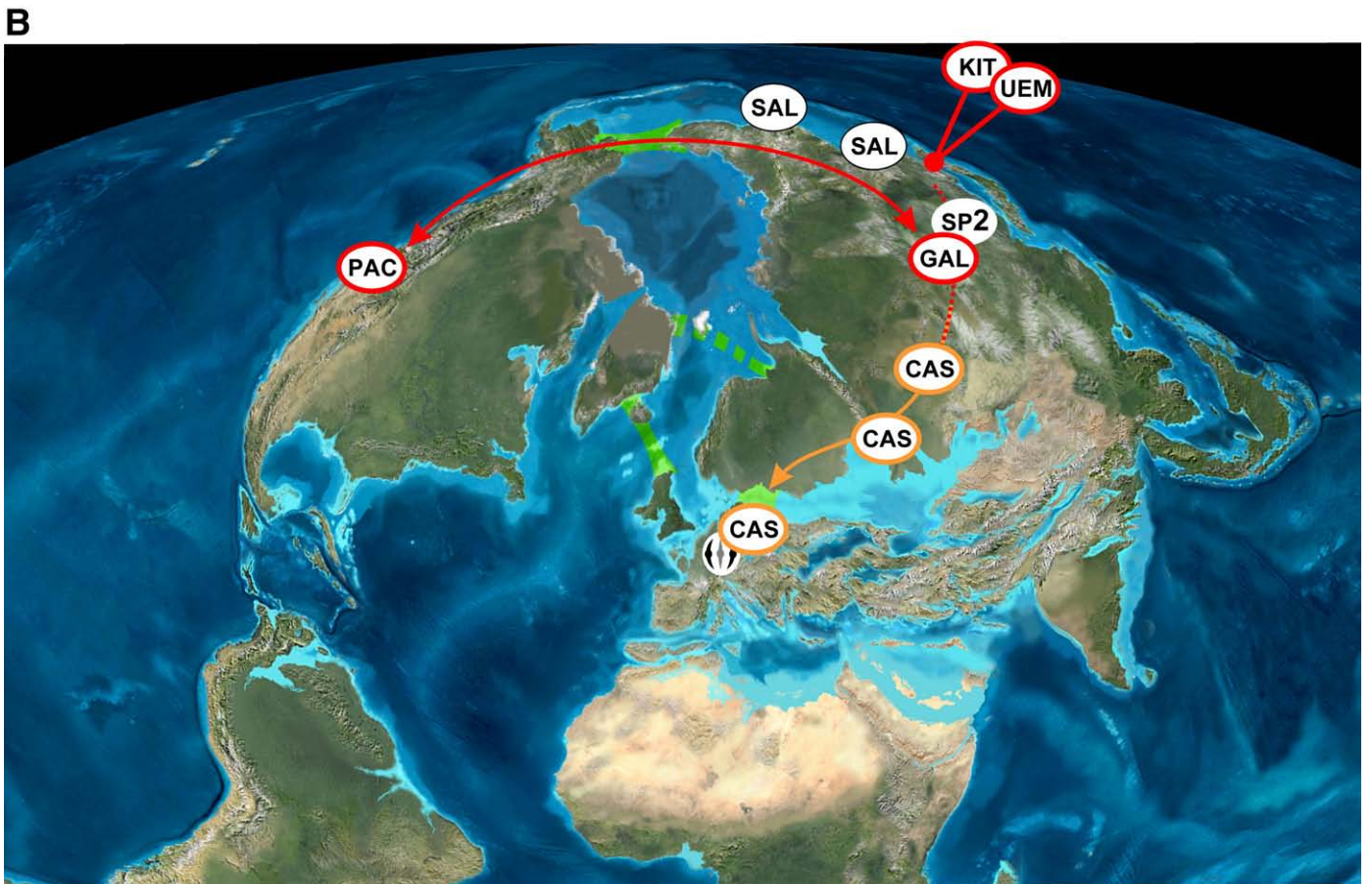
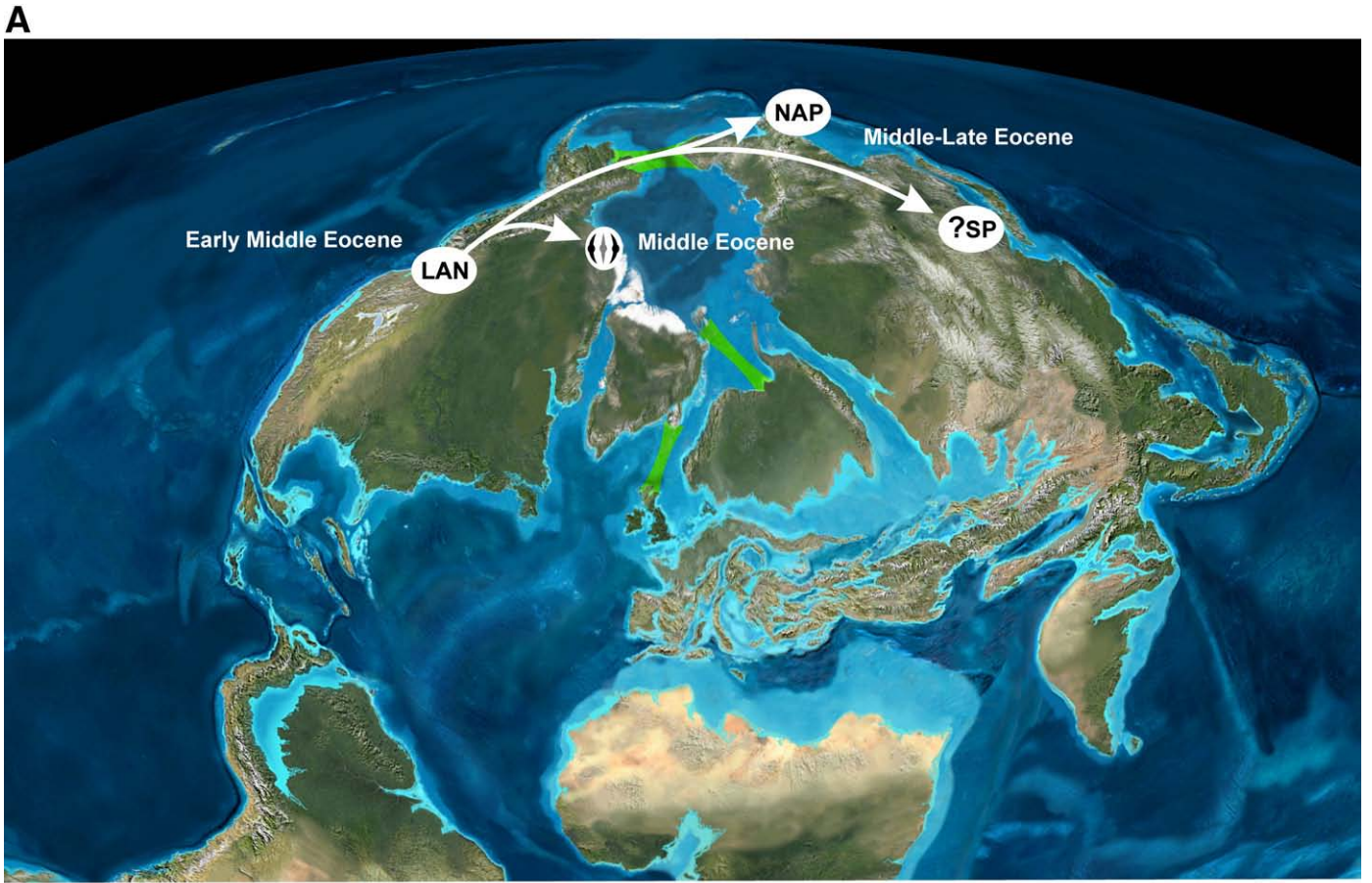
4.1. A comprehensive historical biogeography of *Fagus*

A synthesis of the historical biogeography of the genus *Fagus* is presented in Fig. 10. An origin of *Fagus* in the North Pacific region is likely (Fig. 10A). No remains of *Fagus* have been recovered from the Palaeogene of Ellesmere Island, Greenland, Spitsbergen, Scotland, and Denmark (Manum, 1962; Koch, 1963; Kvaček et al., 1994) pointing to the absence of *Fagus* in the North Atlantic area at that time. Nevertheless, well-preserved dispersed pollen of *Fagus* is found in Middle Eocene sediments at high palaeolatitudes in North America (Axel Heiberg Island; McIntyre, 1991; Figs. 3, 10A), where it constituted an accessory element of rich deciduous conifer and broad-leaved forests (Jahren, 2007). Since both the North Atlantic and the North Pacific areas have been extensively studied palaeobotanically, the pattern appears not to be biased by different quality and amount of sampling in the two regions. This provides substantial evidence for a northern Pacific Eocene radiation of the genus. An out-of-China/East Asia scenario (one of the possibilities indicated by DIVA analysis of extant taxa, Fig. 4) is unlikely in the light of the fossil record (Figs. 3, 5–9). In *Fagus*, the present peak of biodiversity in China can be explained by the young relictual character of the Chinese range of *Fagus* indicating several migrations from Central Asia and north-eastern Asia (Fig. 10D).

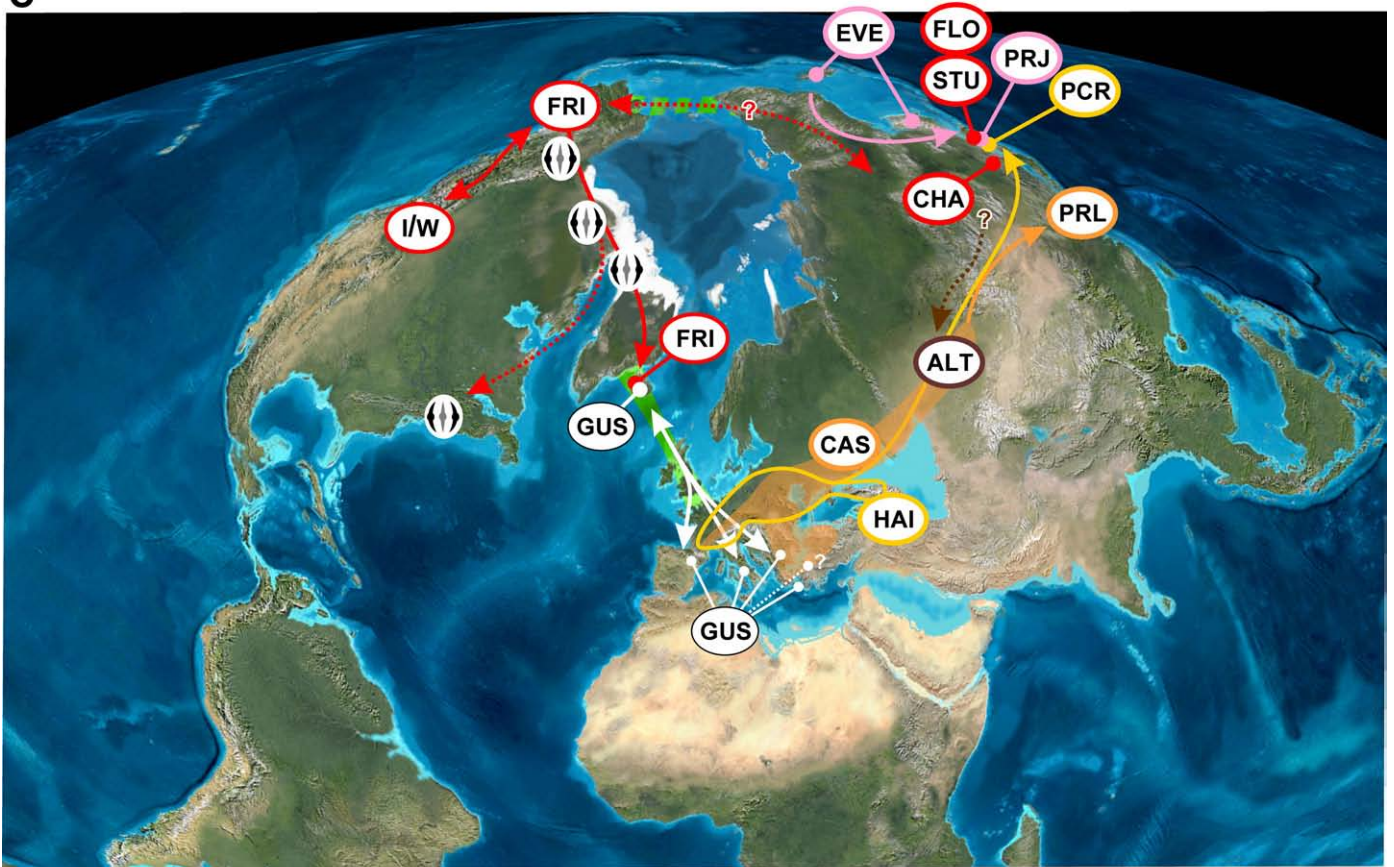
Fagus langevinii represents the first known population, and possibly the common ancestor or a close relative of the common ancestor (Figs. 5–9). Permineralized *Fagus* nuts from the Eocene of Vancouver Island differ from the nuts of *F. langevinii* by the absence of wings (Mindell et al., 2009). This demonstrates either an ancestral polymorphism among the first *Fagus* (also found in several modern species; Shen, 1992; Denk and Meller, 2001; Electronic Supplement) or the existence of two distinct species. At about the same time, *Fagus* is known from Kamchatka and northeastern China (Fotjanova, 1982; Fushun coal mine, unpublished data, images provided by S. R. Manchester).

During the Eocene to Oligocene, NEA (in the Oligocene), BER, and PNA formed a homogeneous biota facilitating the unhindered dispersal of the genus (Fig. 10A,B). From the North Pacific region the genus expanded its range into CAS and EWA in the Oligocene (Fig. 10B). Also CAS and EWA appear to have formed a homogeneous biota from the Late Oligocene to the Middle Miocene. This range expansion was linked to the first major radiation of the genus, as exemplarily illustrated by *F. castaneifolia* (orange signature, Fig. 10B)

Fig. 10. Historical biogeography of beech trees based on the fossil record and in the light of morphological and molecular data. Palaeomaps (A, 50 Ma; B, 35 Ma; C, 20 Ma; D, Present) are Mollweide views, projected through the prime meridian and tilted 45° N (courtesy R. Blakey, July 2009). Active land bridges indicated in light green; pollen symbol refers to dispersed *Fagus* pollen. A, Eocene, initial radiation. LAN, *F. langevinii*; NAP, *F. napanensis*; ?SP, *Fagus* sp. from Fushun, northeastern China. B, Oligocene, first differentiation into a Pacific lineage (red) and continental Eurasian lineage (orange). PAC, *F. pacifica*; GAL, *F. galbanifolia*; KIT, *F. kitamensis*; UEM, *F. uemurae*; CAS, *F. castaneifolia*; Taxa of unknown affinity: SAL, *F. salnikovii*; SP2, *Fagus* sp. 2. C, Miocene, distribution and diversity maximum. New lineages represent ancestors and/or relatives of subgenus *Engleriana* (pink signature; EVE, *F. evenensis*, PRJ, *F. protojaponica*), *F. chienii-lucida* (brown; ALT, *F. altaensis*) and *F. crenata-sylvatica* (yellow; HAI, *F. haidingeri*; PCR, *F. palaeocrenata*), *F. idahoensis-washoensis*; FRI, *F. friedrichii*; CHA, *F. chankaica*; FLO, *F. florinii*; STU, *F. stuxbergii*; PRI, *F. protolongipetiolata*. The taxonomic affinity of *F. gussonii* (GUS) is unknown. D, Pliocene to Present, shaping of modern areas. GRA, *F. grandifolia*; ENG, *F. engleriana*; JAP, *F. japonica*; OKA, *F. okamotoi*; HAY, *F. hayatae*; LON, *F. longipetiolata*; CHI, *F. chienii*; LUC, *F. lucida*; CRE, *F. crenata*; SYL, *F. sylvatica*.



C



D

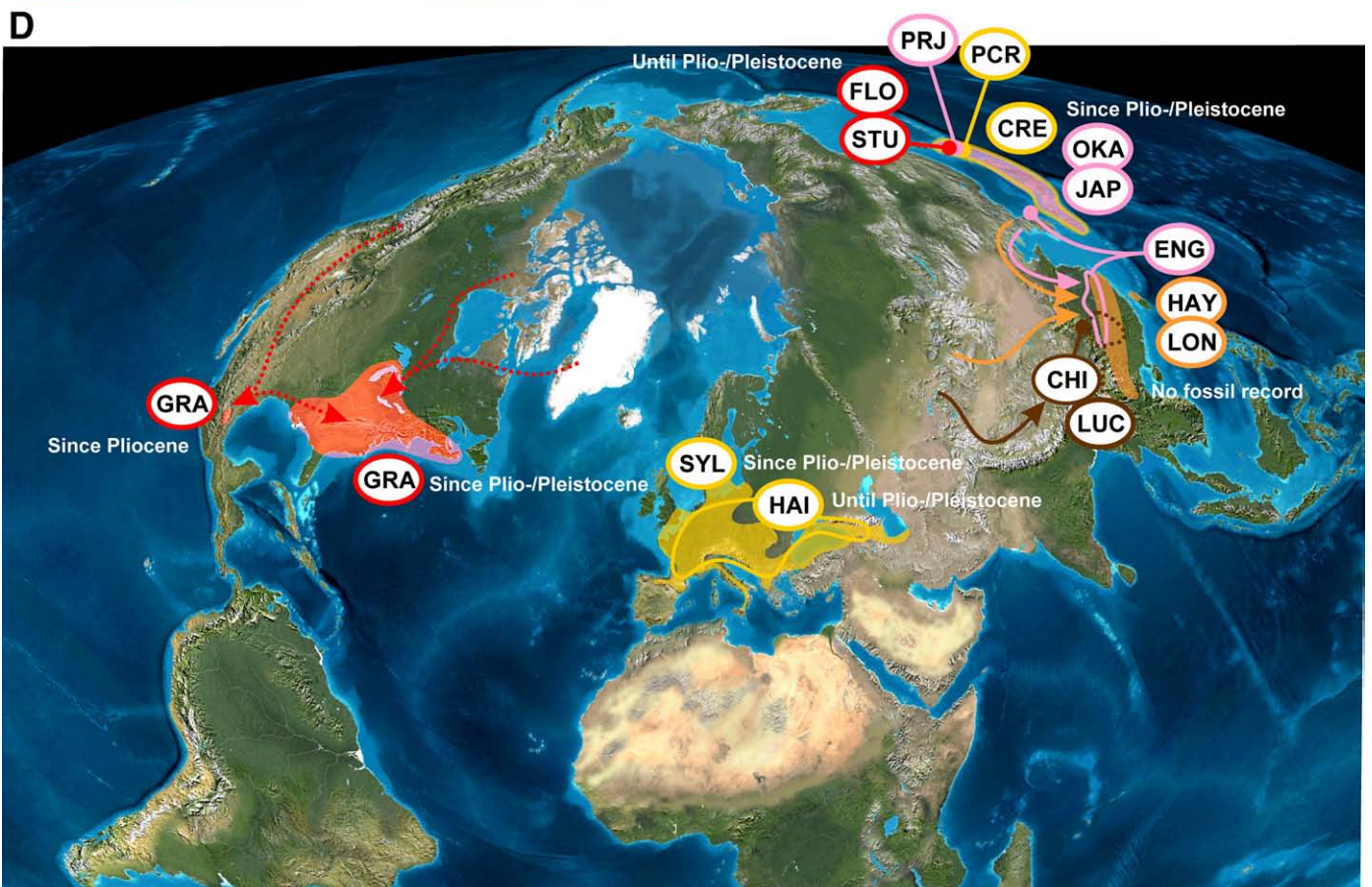


Fig. 10 (continued).

and *F. pacifica* and allies (Figs. 7–9; Denk et al., 2005; indicated by red signatures in Fig. 10B).

A second phase of radiation started in the Late Oligocene–Early Miocene. At this time subgenus *Engleriana* emerged (*F. evenensis*; pink signature, Fig. 10C) and morphological features evolved that are strongly conserved and confined within this lineage (cf. Shen, 1992; Denk, 2003). *Fagus pacifica*-related types became extinct in Northeast Asia in the course of the Miocene and at the latest in the Pliocene, but only after this lineage had reached Iceland (*F. friedrichii*; Fig. 10C). In this case, Iceland was probably colonized from Beringia via North America rather than from Europe. A northern link, Beringia–Iceland corridor, is suggested by Early to Middle Miocene leaf fossils and dispersed pollen from Alaska, Yukon, Banks Island, and possibly Devon Island (Fig. 10C; Whitlock and Dawson, 1990; Fyles et al., 1994; White and Ager, 1994; Grímsson and Denk, 2005). In the Middle and Upper Miocene, first differentiations occurred within the Eurasian populations gradually isolating ancestors of modern lineages (*F. sylvatica* + *F. crenata*; *F. lucida* + *F. chienii*, *F. hayatae* + *F. longipetiolata*; cf. Denk, 2003; Denk et al., 2005; yellow, brown, and orange signatures, Fig. 10C, D). In Europe the differentiation was accompanied by a gradual change of *F. castaneifolia* morphotypes into *F. haidingeri* and possibly *F. gussonii* morphotypes (Kvaček and Walther, 1991; Walther, 1994) in the course of which particular characteristics of *F. castaneifolia* were lost such as narrow trullate leaves with simple dentition, high number of secondary veins, and small stomata (reviewed in Denk, 2004; Electronic Supplement). The Miocene–Pliocene *Fagus haidingeri* is characterized in Central and eastern Europe by the combination of morphotypes indistinguishable from and possibly leading to the modern lineage of *F. crenata* and *F. sylvatica* (ovate leaf shape in combination with distinctly crenate leaf margin; cf. Denk, 2004; Fig. 6; yellow signature, Fig. 10C,D) and morphotypes resembling the Chinese (including Taiwan) *F. hayatae* and *F. longipetiolata* (trullate leaves with distinct tooth morphology, often subsidiary teeth, cf. Denk, 2004; Figs. 6, 7, 9). Also, *F. haidingeri* is the only fossil species with relatively larger pollen, which is characteristic of all modern members of *Fagus* subgenus *Fagus* except for *F. hayatae* and possibly *F. lucida* (Denk, 2003; T. Denk, pers. obs. for *F. lucida*). Iceland and Europe were connected at that time, as is shown by *F. gussonii* (white arrows; Fig. 10C). The origin, direction of migration, and taxonomic affinities of *F. gussonii* are unclear (Figs. 5–8; Fig. 10C), however, *F. gussonii* is unique in combining leaf morphotypes that are identical to variants found in *F. sylvatica* from Turkey and *F. longipetiolata* (Denk, 2004). It is also more similar to the older continental Eurasian *F. castaneifolia* than to the North American–Icelandic *F. friedrichii* (Fig. 9C,D).

The Central Asian portion of *F. castaneifolia* is most likely the stock or one stock from which *F. hayatae* and *F. longipetiolata* evolved in post-Miocene time (Fig. 9C, E, F; orange signature; Fig. 10D). Shared characters include a high leaf index, trullate leaf shape, small stomata (in *F. hayatae* and *F. castaneifolia*), veinlets of areoles always simple, and spine-like appendages of cupules (Denk, 2003; Electronic Supplement). Due to the generally plesiomorphic appearance of the extant species *F. hayatae* and *F. longipetiolata* (Denk, 2003; Denk et al., 2005; Figs. 5–9), the time of their origin can hardly be established, and it is impossible to determine which of the Middle and Upper Miocene fossils could represent (additional) ancestors of *F. hayatae*–*longipetiolata* (e.g. *F. florinii*; Fig. 9E, F). The conspicuous intraindividual ITS polymorphism found in *F. hayatae* and *F. longipetiolata* (Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008) could be indicative for more than one ancestral lineage that was involved in the formation of these two species (Fig. 9E, F). *Fagus altaensis* indicates a fragmentation of the continental Eurasian range during the Miocene, allowing speciation (brown lines, Fig. 10) manifested in detectable morphological change (Figs. 5–9). The (sister) relationship between *F. altaensis* and *F. lucida* indicated by the analysis (Figs. 5–7) may either reflect a direct relationship (brown signature, Fig. 10C, D) or long branch attraction (due to their general dissimilarity to all other

taxa, Figs. 7–9; Electronic Supplement). The characteristics of the cupule (appendages small scales), however, may link *F. altaensis* to *F. lucida* (Iljinskaya, 1982).

In a last phase, ancestors of *F. engleriana* migrated into China in the latest Neogene, and became extinct in most parts of Northeast Asia, except for Ullung Do (South Korean island) and Japan. They met with the ancestors of *F. lucida* and *F. hayatae*–*longipetiolata* that migrated to Southern China from Central Asia and possibly continental Northeast Asia (Fig. 10D). These lineages at that time must have been well isolated considering the distinctiveness of *F. lucida* and *F. engleriana* (Figs. 5–7; Denk, 2003; Grimm, 2003; Denk et al., 2005; Grimm et al., 2007b). In western Eurasia *F. haidingeri* persisted into the Pliocene and rarely into the Pleistocene (eastern Georgia; T. Denk, pers. obs.); Ancestral morphotypes within *F. haidingeri*, resembling modern Chinese species, were subsequently lost and only the modern types (equalling *F. sylvatica*; see above) survived. It is not yet entirely clear which role *F. gussonii* played in the formation of *F. sylvatica* (cf. Figs. 5–7; Fig. 10C). The separation between *F. crenata* and *F. sylvatica* must have started with the fragmentation of EWA and NEA after CAS had become an unsuitable habitat for *Fagus* in the course of the Miocene (Ramstein et al., 1997; Fig. 10C). A later link between EWA and NEA is highly improbable. The leaf fossil *F. palaeocrenata* (Miocene to Pleistocene of Japan) and patterns of intraindividual ITS variability (Grimm et al., 2007b; Göker and Grimm, 2008) may reflect a Miocene divergence between the lineages leading to *F. sylvatica* and *F. crenata*. The biogeographic and phylogenetic history of *F. chienii* remains obscure (cf. Fig. 5 vs. Figs. 6–9; Fig. 10D; Electronic Supplement).

4.2. How to reconstruct biogeographic history

Any analysis that uses the recent presence or absence of a taxon or lineage in a certain area as a character for cladistic analysis is fundamentally flawed (Ronquist, 1994, 1996; Santos, 2007). In contrast, process-based analyses, such as DIVA, principally are more appropriate, but still are hampered by missing data and methodological limitations (Ronquist, 1996, 1997; this study). For a modern group with a disjunct Northern Hemispheric distribution such as *Fagus* DIVA analyses are uninformative (Fig. 4) for the reasons given in Ronquist (1996). Inclusion of outgroups such as other Fagaceae or *Nothofagus* does not add better resolution as these groups are similarly or even more disjunct resulting in equivocal points of origin for them (details not shown). Moreover, they may indicate a wrong root (Fig. 1B). Given a good fossil record as in *Fagus*, a narrative (documentary) approach is likely to perform better: Changing areas, migration pathways, and extinction events are known to and simultaneously considered by the narrator (this study, Fig. 3; Denk, 2004; see also Felsenstein, 2004) along with a deep-as-possible understanding of evolutionary relationships (Figs. 5–9; Denk et al., 2005; see also Mayr and Bock, 2002). As emphasized by Mayr and Bock (2002), cladograms or any other phylogenetic reconstruction, can only provide working hypotheses, but a deep knowledge of the group under consideration is essential. For instance, the information provided in Figs. 5–9 can only be fully extracted and properly discussed in the light of the spatiotemporal unfolding of *Fagus* (Fig. 3) and taxonomic (morphological and molecular) expertise on fossil and extant taxa (Denk et al., 2002; Grimm, 2003; Denk, 2003, 2004; Denk et al., 2005).

Moreover, the naïve belief of many users of cladograms that a cladogram can be directly translated in an evolutionary history is far from the reality of most if not all real-life data sets (see fig. 1.11 in Kitching et al., 1998). In *Fagus*, a single cladogram cannot possibly reflect the evolution of the genus (Fig. 2; Denk, 2003; Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008).

Another inherent problem is that the placement of the root in phylogenetic trees is always problematic (Felsenstein, 2004). If done by knowledge as shown in Fig. 3, using basal dichotomy (BDICH) instead of the reconstructed basal Eurasian grade (BEAG) topology (see Denk, 2003), it can be criticized as subjective, if done cladistically

(outgroup rooting), it is circular reasoning (Mayr and Bock, 2002; see also Fig. 1). But, knowing the correct root is a prerequisite of any biogeographic reconstruction based on input trees. If the root is not known, all alternative rootings would need to be considered (Ronquist, 1996), or tests would need to be applied about which root is the best. In practice, however, this is never done. In *Fagus*, such a test is impossible due to the complex genetic differentiation seen among modern members of the genus (Denk et al., 2005; Grimm et al., 2007b; Göker and Grimm, 2008).

Moreover, including fossils in phylogenetic analyses significantly increases, for instance, the number of MPT (Wiens, 2003), and therefore the number of cladograms that would have to be considered in a biogeographic analysis combining fossils and extant taxa; in our case, 105 MPT that are topologically largely incongruent (Fig. 5). Sophisticated theoretical approaches such as ML-based reconstructions of biogeography (Ree et al., 2005; Sanmartin et al., 2008) may successfully surpass primary limitations of DIVA and earlier approaches as outlined before, but they still use rooted phylograms (or chronograms) based on extant taxa as the sole source of input data. Any systematic error induced by the input phylogram, for instance under- or unrepresented areas/taxa (such as Europe, Central Asia, Beringia, Pacific North America, and Iceland in the case of *Fagus*), will result in an erroneous reconstruction but will remain undetected. Many woody angiosperm taxa were important in Pacific North America during the Palaeogene but later went extinct in this region (*Actinidia* [Actinidiaceae], *Ailanthus* [Simaroubaceae], *Anamirta* [Menispermaceae], *Craigia*, *Tilia* [Malvaceae: Tilioideae], *Cyclocarya*, *Pterocarya* [Juglandaceae], *Dipteronia* [Sapindaceae], *Davidia* [Cornaceae], *Emmenopterys* [Rubiaceae], *Ensete* [Musaaceae], *Iodes* [Icacinaeae], *Sargentodoxa* [Sargentodoxaceae], *Tapiscia* [Tapisciaceae], *Toricellia* [Toricelliaceae], *Trochodendron* [Trochodendraceae]; Manchester, 1987, 1999). Biogeographic reconstructions (scenarios) based on extant phylogenies need to be tested against the facts provided by the fossil record. If fossils can be found that fit into a neo-biogeographic scenario, they provide the only proof that, at least some areas, have been correctly reconstructed. If the fossil record is at odds with the biogeographic scenario inferred from extant taxa, this may have two reasons: (1) the fossils may be misinterpreted, (2) the reconstruction may be wrong, for example because of incorrect input tree(s). Contrary to what is often assumed, the latter cannot be overcome by adding more molecular data (e.g., Delsuc et al., 2005). In contrast, each effort put into recovering plant bearing sediments and thorough descriptions of fossil plants directly result in a better (enhanced) biogeographic framework. For instance, *F. langevinii* demonstrates that the genus *Fagus* occurred in the Eocene of western North America. *Fagus friedrichii* demonstrates that there was a link between Iceland and Alaska during the Miocene. *Fagus castaneifolia* demonstrates a widespread Oligocene to Miocene distribution of *Fagus* across Eurasia and provides the basis to understand modern patterns of morphological, molecular and geographic differentiation. If *Fagus* remains would be recovered in Paleocene sediments of Spitsbergen, for instance, the current hypothesis of a Northern Pacific origin of *Fagus* would have to be changed. Such a modification could never be possibly achieved by extant data sets (adding more genes, more comprehensive sampling; more sophisticated methods of inference).

5. Conclusions

‘Narrative’ (palaeontologically documented) biogeography is historical in the sense that it is based on plant fossils and their stratigraphic, taxonomic, and phylogenetic position, and on the tectonic evolution of the earth. It is neither worse, nor should it be seen in opposition to numerical (algorithmal) biogeography. Biogeographical analyses require fossils and their comprehensive understanding in order to be testable. If numerical biogeographical studies based on extant taxa fail to reconstruct the historical biogeography of a small group such as *Fagus*, they may also result in erroneous

reconstructions for larger groups lacking a fossil record. Any reconstruction of biogeography remains purely hypothetical as long as it is not validated against external data sets, such as the fossil record and evidence from plate tectonics.

A holistic approach to historical biogeography integrating narrative biogeography (Fig. 3), phylogenetic reconstructions (Figs. 5–7), and investigation of overall similarity (Figs. 7–9) as proposed here is much more time consuming and therefore it is less practiced (cf. Tiffney and Manchester, 2001; Xiang et al., 2006). However, it is the only means to reconstruct a historical biogeography that is in accordance, and not in contrast, to the facts of the geological past (Fig. 10).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.revpalbo.2009.08.007.

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